NAS JIAXING · PROFESSIONAL DEVELOPMENT

AI for Education Initiative

Building Local AI Infrastructure for Teachers and Students

Prepared by William Morris | May 2026

Executive Summary

We propose establishing a dedicated AI education server at NAS Jiaxing — a local, privacy-focused platform that gives teachers and students direct access to cutting-edge AI tools without relying on external cloud services or VPNs.

The Problem We Are Solving

Teachers want to use AI in classrooms, but face three barriers:

1. **Access blocked** — Many AI tools (ChatGPT, Claude, Gemini) are blocked on school networks without VPN
2. **Data privacy** — Uploading student work to foreign cloud services raises compliance concerns
3. **Cost and complexity** — Premium AI subscriptions are expensive; teachers don't know where to start

Our Solution

A local AI server running inside the school network, hosting open-source AI models and educational tools that are:

- **Always accessible** — No VPN needed; works on school WiFi
- **Fully private** — All data stays inside the school; no external cloud uploads
- **Free to use** — Open-source software; no subscription fees
- **Educational** — Designed specifically for teaching, not general chat

What Is Already Working (Demo)

We have built a proof-of-concept running on a personal workstation. This demonstrates that the technology works and the concept is viable.

Current Demo Machine Specs

Component	Specification
CPU	AMD Ryzen 9 7950X3D (16 cores, 32 threads)

RAM	96GB DDR5
GPU	AMD RX 7900 XTX (24GB VRAM)
Storage	2TB NVMe SSD + 4TB HDD
OS	Ubuntu 24.04 LTS
Network	School LAN — accessible from any classroom
Total Build Cost	~¥20,000 (personal funds)

This machine currently serves educational content and AI tools to the school network. It proves the concept. For a permanent school solution, we recommend a purpose-built server at roughly half the cost.

Proposed School Server — Two Build Options

For a permanent, school-owned installation, we researched two viable approaches at the same ~¥10,000 budget. Both are significantly more cost-effective than the ¥20,000 demo build.

⭐ Recommended: Used Enterprise Server (Option A)

A professional 2U rackmount server with 32-core AMD EPYC processor. Enterprise-grade reliability with redundant power, remote management, and massive expansion capacity. This is what real data centers use — repurposed for education.

Option A: Used Enterprise Server (~¥10,100) ⭐

Component	Specification	Est. Price
Chassis	GIGABYTE G292-Z20 2U Rackmount Server	¥4,700
Motherboard	MZ22-G20 (SP3 socket, 8x DDR4, 10x PCIe x16)	
CPU	AMD EPYC 7532 — 32 cores / 64 threads, 2.4GHz, 256MB L3 cache	
RAM	64GB DDR4 ECC RDIMM (8-channel, expandable to 512GB)	¥1,000
GPU	AMD RX 7800 XT (16GB VRAM) or reuse demo 7900 XTX	¥3,500
Storage	1TB NVMe SSD + 4TB HDD (8x 2.5" hot-swap bays)	¥900
Power	2200W Dual Redundant PSU (enterprise-grade, hot-swap)	Included
Cooling	4x Server-grade fans with thermal management	Included
OS & Software	Ubuntu 24.04 LTS (free, open-source)	¥0
Total	**32 cores · 64GB RAM · 16GB VRAM · Rackmount**	**~¥10,100**

Option B: New Desktop Build (~¥10,200)

Component	Specification	Est. Price
CPU	AMD Ryzen 7 7700X (8 cores, 16 threads)	¥2,000
RAM	64GB DDR5 (expandable to 128GB)	¥2,500
GPU	AMD RX 7800 XT (16GB VRAM)	¥3,500
Storage	1TB NVMe SSD + 4TB HDD	¥1,200
Case/PSU/Motherboard	Standard ATX mid-tower, 750W PSU	¥1,500
OS & Software	Ubuntu 24.04 LTS (free, open-source)	¥0

Total	**8 cores · 64GB RAM · 16GB VRAM · Desktop**	**~¥10,200**

Side-by-Side Comparison

Feature	Option A: Used Enterprise	Option B: New Desktop
CPU Cores	◍ 32 cores / 64 threads	8 cores / 16 threads
Form Factor	◍ 2U Rackmount (fits server room)	Mid-tower desktop (sits on floor)
Power Supply	◍ Dual 2200W redundant (hot-swap)	Single 750W (no redundancy)
PCIe Expansion	◍ 10x PCIe x16 slots (add GPUs/storage later)	3-4 slots (limited)
Drive Bays	◍ 8x hot-swap 2.5" bays	2-3 internal bays
Remote Management	◍ IPMI/iKVM (reboot/fix remotely)	None (need physical access)
Warranty	1 year seller warranty	◍ 3-year component warranties
Power Efficiency	200W TDP (enterprise optimized)	◍ Lower idle power (105W TDP)
Noise	Louder (server fans)	◍ Quieter (desktop fans)
Memory Type	DDR4 ECC (cheaper to expand)	◍ DDR5 (newer, faster)
Vendor Lock-in	None (standard AMD platform)	None (standard AMD platform)

Why We Recommend Option A (Used Enterprise Server)

For a school environment, the used enterprise server is the better long-term investment:

- **4x more CPU cores** — handles more simultaneous users and larger AI models
- **Rackmount form factor** — sits in the school server room, not under a desk
- **Redundant power** — if one PSU fails, the other keeps running. Zero downtime for students.
- **Massive expansion** — 10 PCIe slots means we can add more GPUs, network cards, or storage later without replacing the whole machine
- **Remote management** — IT can reboot, diagnose, or reinstall remotely without walking to the machine
- **Enterprise reliability** — these servers run 24/7 for years in data centers. A desktop PC isn't designed for that.

The only trade-off: It's slightly louder and uses DDR4 instead of DDR5. For a server room, noise doesn't matter. And DDR4 ECC is cheaper to expand later (up to 512GB).

What the Server Will Host

For Teachers — Professional Development

- **AI Lesson Plan Assistant** — Generate differentiated activities, rubrics, and assessments

- **Text Simplifier** — Adapt reading passages for ESL students by level
- **Discussion Starter Generator** — Create engaging prompts for any topic
- **Vocabulary Builder** — Auto-generate word lists with definitions and example sentences
- **Reading Assessment Creator** — Build comprehension questions with answer keys

For Students — Learning Tools

- **Educational Games** — Grammar, vocabulary, and critical thinking games
- **Creative Project Generator** — AI-assisted story writing and project ideas
- **AI Tutor** — Students can ask questions and get explanations without leaving the school network
- **Listening Practice** — AI-generated listening exercises with transcripts

For the School — Infrastructure

- **Internal AI Resource Website** — `ai.nasjiaxing.cn` — a professional portal for all AI tools
- **Content Management** — IT controls what content is available; everything is curated for education
- **No External Dependencies** — Works even if the school's internet connection is slow or filtered

Benefits to NAS Jiaxing

For Teachers

- Save hours on lesson planning and material creation
- Differentiate instruction automatically
- Access PD workshop materials anytime
- No need for personal VPNs or subscriptions
- Data stays inside the school — fully compliant

For Students

- Personalized learning tools available 24/7 on campus
- Interactive games that reinforce curriculum
- AI writing and creativity support
- Safe environment — no ads, no tracking, no external data sharing

For the School

- Position NAS Jiaxing as a leader in AI education
- One-time hardware cost vs. ongoing subscription fees
- Full control over content and access
- Scalable — add more capabilities as needed

For IT & Compliance

- All traffic stays on the school LAN
- No student data leaves the building
- Can be placed under school's existing ICP备案
- Centrally managed and monitored

Implementation Timeline

Phase	Duration	Actions
Phase 1: Approval & Procurement	2–3 weeks	Principal and IT approve budget; order server hardware
Phase 2: Setup & Configuration	1 week	Install Ubuntu, configure AI software, set up internal website
Phase 3: Content & Testing	1–2 weeks	Load educational tools, test with pilot teachers, gather feedback
Phase 4: Launch & Training	1 week	Internal launch, PD workshop for staff, onboarding materials
Phase 5: Expansion	Ongoing	Add new tools based on teacher requests, student feedback

Compliance & Security

We are committed to full compliance with Chinese regulations:

- **ICP备案** — The site will operate under the school's existing ICP filing as a subdomain (`ai.nasjiaxing.cn`)
- **Data Sovereignty** — All data stored locally; nothing sent to foreign servers
- **Content Control** — IT and school leadership review and approve all content before go-live
- **No VPN Required** — Everything runs on the school internal network
- **Firewall Friendly** — Standard HTTP on port 80; no special ports or protocols needed

What We Need From the School

📋 Three Requests to Move Forward

💰 **1. Budget Approval (~¥10,000)**

For a dedicated school AI server. We can source used enterprise hardware to stay within budget.

🌐 **2. IT Support — Static IP & Subdomain**

A reserved static IP on the school LAN for the server, and a DNS entry for `ai.nasjiaxing.cn` pointing to that IP.

✅ **3. ICP Subdomain Authorization**

Permission to add `ai.nasjiaxing.cn` under the school's existing ICP备案.

About the Demo

A live demo is already running on the school network. You can access it from any device connected to the school WiFi at:

`http://10.39.26.217/`

Accessible only from the school LAN

This demo shows the landing page, educational games, AI tools, and the general look and feel of what the final site would be.

About Me — William Morris

I am an English teacher at NAS Jiaxing with a background in technology and a personal interest in AI. I built the demo machine in my own time using personal funds to prove that this is feasible and valuable for our school. I am not asking the school to fund my personal project — I am proposing a separate, school-owned server that serves the entire faculty and student body.

I am happy to manage the technical setup, maintain the server, and train teachers on how to use the tools. My goal is to make NAS Jiaxing a model school for AI-integrated education.

Questions? Contact William Morris at william.morris@nasjiaxing.cn

Full technical details available upon request · Demo accessible on school network